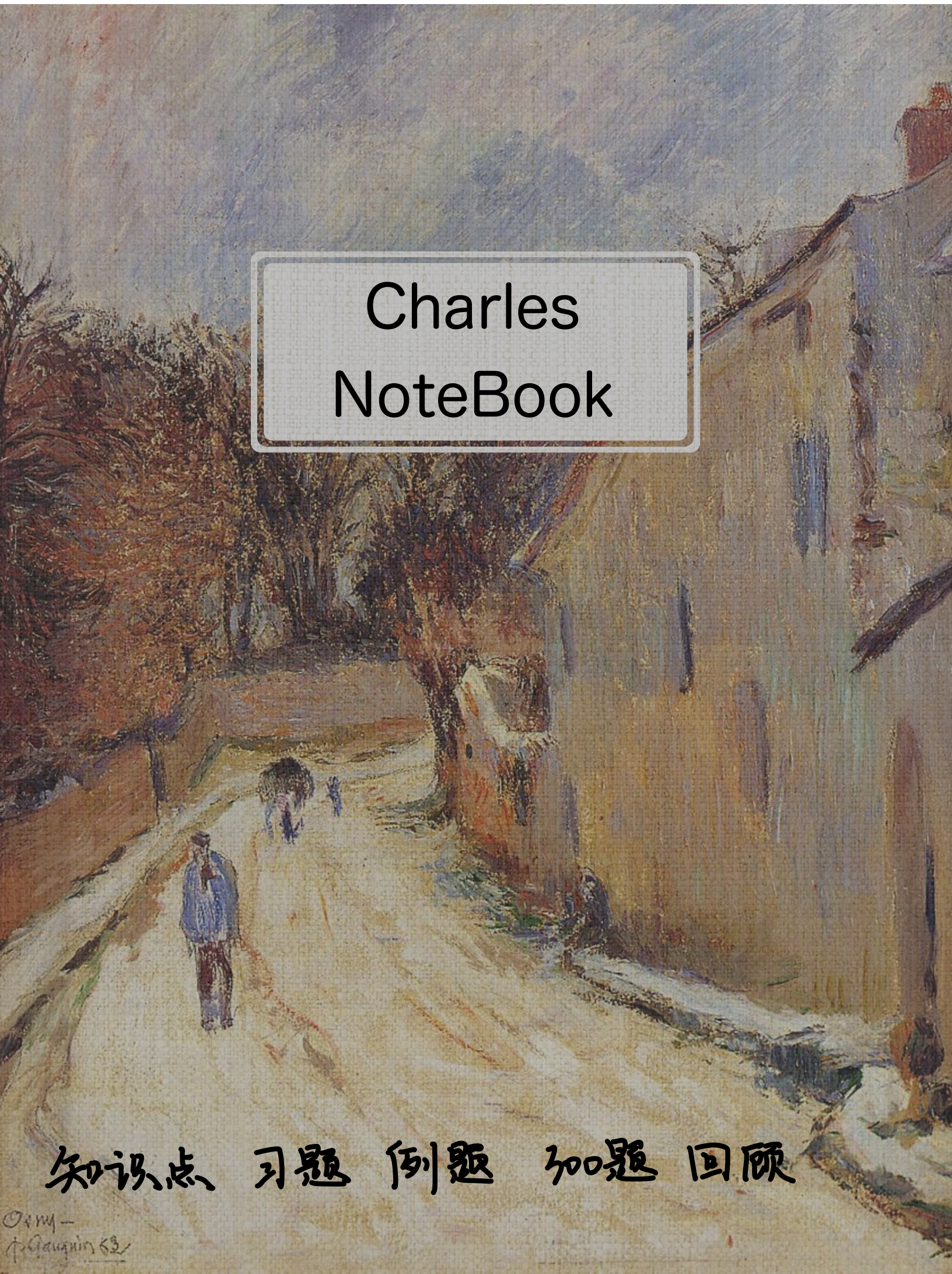


An impressionist painting of a street scene. On the left, there are large, leafy trees in shades of brown and orange. A dirt path leads from the foreground into the distance. Several figures are walking along the path: a man in a blue coat and hat in the foreground, and a group further back. On the right, there is a tall, light-colored building with a red roof. The sky is a mix of blue and white, suggesting a bright, slightly overcast day. The overall style is loose and painterly, with visible brushstrokes.

Charles NoteBook

知识点 习题 例题 加题 回顾

- 1 知识点
- 2 习题
- 3 300题

函数极限

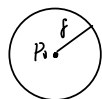
一. 函数极限

- 邻域 & ε 语言 (四句话)
- 等式脱帽法
- 泰勒公式

$$\begin{aligned}\sin x &= x - \frac{1}{6}x^3 + o(x^3) \\ \arcsin x &= x + \frac{1}{6}x^3 + o(x^3) \\ \tan x &= x + \frac{1}{3}x^3 + o(x^3) \\ \arctan x &= x - \frac{1}{3}x^3 + o(x^3)\end{aligned}$$



① 邻域 $(x_0 - \delta, x_0 + \delta)$
↓
"附近"



$$U(P_0, \delta) = \{(x, y) \mid \sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta\}$$

② 函数极限的定义 —— 去心邻域内有定义 (与 P_0 点无关)

四句话 $\lim_{x \rightarrow x_0} f(x) = A$: $\forall \varepsilon > 0, \exists \delta > 0, |x - x_0| < \delta, \text{ s.t. } |f(x) - A| < \varepsilon$

$$\begin{aligned}\text{注1. } x \rightarrow x_0 & \quad \exists \delta > 0, |x - x_0| < \delta, x \in (x_0 - \delta, x_0 + \delta) \\ x \rightarrow x_0^+ & \quad \exists \delta > 0, x - x_0 < \delta, x \in (x_0, x_0 + \delta) \\ x \rightarrow x_0^- & \quad \exists \delta > 0, x_0 - x < \delta, x \in (x_0 - \delta, x_0) \\ x \rightarrow \infty & \quad \exists X > 0, |x| > X \\ x \rightarrow +\infty & \quad \exists X > 0, x > X \quad x > 0 \\ x \rightarrow -\infty & \quad \exists X > 0, -x > X \quad x < 0\end{aligned}$$

[练习] $\lim_{x \rightarrow \infty} f(x) = \infty \iff \forall M > 0, \exists X > 0, x > X, |f(x)| > M$

注2 $A = \lim_{x \rightarrow x_0} f(x) \iff \lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0^-} f(x) = A$

等式脱帽法 $\lim_{x \rightarrow x_0} f(x) = A \iff f(x) = A + \alpha(x), \lim_{x \rightarrow x_0} \alpha(x) = 0$

例 3.16

已知 $\lim_{x \rightarrow 0} \frac{\tan 2x + x f(x)}{\sin x^3} = 0$, 则 $\lim_{x \rightarrow 0} \frac{2 + f(x)}{x^2} = ?$ $x - \sin x \sim \frac{x^3}{6}$

$$\lim_{x \rightarrow 0} \frac{\tan 2x + x f(x)}{\sin x^3} = \lim_{x \rightarrow 0} \frac{\tan 2x + x f(x)}{x^3} = \lim_{x \rightarrow 0} \frac{2x + \frac{(2x)^3}{3} + x f(x)}{x^3} = 0$$

$$\Rightarrow f(x) = -2 - \frac{8}{3}x^2 \quad (x \rightarrow 0)$$

$$\lim_{x \rightarrow 0} \frac{2 + f(x)}{x^2} = \lim_{x \rightarrow 0} \frac{2 - 2 - \frac{8}{3}x^2}{x^2} = -\frac{8}{3}$$

用等式脱帽法: $\lim_{x \rightarrow 0} \frac{\tan 2x + x f(x)}{\sin x^3} = 0 \Rightarrow \frac{\tan 2x + x f(x)}{\sin x^3} = 0 + \alpha(x), \lim_{x \rightarrow 0} \alpha(x) = 0$

$$\Rightarrow f(x) = \frac{\sin x^3 \cdot \alpha(x) - \tan 2x}{x}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{2 + f(x)}{x^2} = \lim_{x \rightarrow 0} \frac{2x + \sin x^3 \cdot \alpha(x) - \tan 2x}{x^3}$$

$$= \lim_{x \rightarrow 0} \frac{2x - \tan 2x}{x^3} + \lim_{x \rightarrow 0} \alpha(x) = -\frac{8}{3}$$

→ 函数极限性质
唯一, 局部有界, 局部保号

→ 极限运算规则

→ 夹逼准则

→ 洛必达法则

② 函数极限性质

唯一性: $\lim_{x \rightarrow \infty} x \rightarrow +\infty$ 且 $x \rightarrow -\infty$

① $\lim_{x \rightarrow \infty} e^x$ 不存在 $\lim_{x \rightarrow +\infty} e^x = +\infty$ $\lim_{x \rightarrow -\infty} e^x = 0$

② $\lim_{x \rightarrow 0} \frac{\sin x}{|x|}$ 不存在 $\lim_{x \rightarrow 0^+} \frac{\sin x}{|x|} = 1$ $\lim_{x \rightarrow 0^-} \frac{\sin x}{|x|} = -1$

③ $\lim_{x \rightarrow \infty} \arctan x$ 不存在 $\lim_{x \rightarrow \infty} \arctan x = \frac{\pi}{2}$ $\lim_{x \rightarrow -\infty} \arctan x = -\frac{\pi}{2}$

④ $\lim_{x \rightarrow 0} [x]$ 不存在 $\lim_{x \rightarrow 0^+} [x] = 0$ $\lim_{x \rightarrow 0^-} [x] = -1$

局部有界性 $\lim_{x \rightarrow x_0} f(x) \exists \rightarrow f(x)$ 有界
eg. $\lim_{x \rightarrow \infty} \sin x$ 不存在

局部保号性 "x $\rightarrow x_0$ " 时 $\lim_{x \rightarrow x_0} f(x) = A > 0 \Rightarrow f(x) > 0$

证明 $\lim_{x \rightarrow x_0} f(x) = A$ $\forall \epsilon > 0, \exists \delta > 0, (x - x_0) < \delta, |f(x) - A| < \epsilon$

取 $\epsilon = \frac{A}{2} > 0 \Rightarrow |f(x) - A| < \frac{A}{2}$

$\Rightarrow f(x) > \frac{A}{2} > 0$

④ 极限运算规则

$\lim_{x \rightarrow x_0} f(x) = A$ $\lim_{x \rightarrow x_0} g(x) = B$ 则 $f(x) \pm g(x)$

⑤ 夹逼准则

$g(x) \leq f(x) \leq h(x)$

$\downarrow \quad \quad \quad \downarrow$
 $A \rightarrow A \leftarrow A$

* $g(x) \leq f(x) \leq h(x)$ 且 $\lim_{x \rightarrow x_0} (h(x) - g(x)) = 0$ 不能推出 $\lim_{x \rightarrow x_0} f(x)$.

反例: 当 $x > 0$ 时, $h(x) = x + \frac{1}{x+1}$, $f(x) = x + \frac{1}{x+1}$, $g(x) = x + \frac{2}{x+1}$

且 $\lim_{x \rightarrow \infty} (h(x) - g(x)) = 0$, 但 $\lim_{x \rightarrow \infty} x + \frac{2}{x+1} = +\infty$

⑥ 洛必达法则

① $x \rightarrow a$ (或 $x \rightarrow \infty$) 时 $f(x)$ 与 $F(x) \rightarrow 0$

② $f(x)$ 与 $F(x)$ 可导

③ $\lim_{x \rightarrow a} \frac{f'(x)}{F'(x)}$ 存在或无穷大

则 $\lim_{x \rightarrow a} \frac{f(x)}{F(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{F'(x)}$

若洛必达后极限不存在, 不能判定原极限不存在

eg $\lim_{x \rightarrow 0} \frac{x^2 \sin \frac{1}{x}}{x} = 0$ 但 $\lim_{x \rightarrow 0} 2x \sin \frac{1}{x} - \cos \frac{1}{x}$ 不存在

→ 泰勒公式

⑦ 泰勒公式

$$\left\{ \begin{array}{l} 1 \quad \left\{ \begin{array}{l} e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + o(x^3) \\ \cos x = 1 - \frac{x^2}{2!} + o(x^2) \\ \sin x = x - \frac{x^3}{3!} + o(x^3) \\ \operatorname{arcsin} x = x + \frac{x^3}{3!} + o(x^3) \end{array} \right. \\ 2 \quad \left\{ \begin{array}{l} \tan x = x + \frac{x^3}{3} + o(x^3) \\ \operatorname{arctan} x = x - \frac{x^3}{3} + o(x^3) \end{array} \right. \\ 3 \quad \left\{ \begin{array}{l} \frac{1}{1-x} = 1 + x + x^2 + x^3 + o(x^3) \\ \frac{1}{1+x} = 1 - x + x^2 - x^3 + o(x^3) \\ \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + o(x^4) \end{array} \right. \\ 4 \quad (1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + o(x^2) \\ \text{如 } \sqrt{1+x} = (1+x)^{\frac{1}{2}} = 1 + \frac{1}{2}x + \frac{\frac{1}{2}(-\frac{1}{2})}{2} x^2 + o(x^2) \end{array} \right.$$

• (2019) $x \rightarrow 0$ 时

$$\begin{aligned} 1 - \frac{x}{2!} + \frac{x^2}{4!} - \dots &= \cos x \\ (1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots) &= \cos x \end{aligned}$$

• $x - \tan x \sim -\frac{x^3}{3}$

$\tan x - \sin x \sim \frac{x^3}{2}$

$\operatorname{arcsin} x - \operatorname{arctan} x \sim \frac{x^3}{2}$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{\operatorname{arctan} x - \operatorname{arcsin} x} = \lim_{x \rightarrow 0} \frac{\frac{x^3}{2}}{-\frac{x^3}{2}} = -1$$

⑧ 高阶无穷小计算规则

无穷小比较 $m, n \in \mathbb{Z}^+$

a $o(x^m) \pm o(x^n) = o(x^{\min(m,n)})$

高阶无穷小 + 低阶无穷小 = 低阶无穷小

b $o(x^m) \cdot o(x^n) = o(x^{m+n})$

c $o(x^m) = o(kx^m) = k \cdot o(x^m), k \neq 0$

A/B 上下同阶原则 eg. $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} = \lim_{x \rightarrow 0} \frac{\frac{1}{6}x^3 + o(x^3)}{x^3} = \frac{1}{6}$

如 A-B 幂次最低原则 eg. $x \rightarrow 0 \quad \tan x - \sin x = (x + \frac{x^3}{3} + o(x^3)) - (x - \frac{x^3}{6} + o(x^3))$
原式不变最低次幂

$x + \sin x \sim 2x$

eg $x \rightarrow 0 \quad x + \sin x = x - (-\sin x)$
 $= x - (-x + o(x)) = 2x + o(x)$

(2021) $\lim_{x \rightarrow 0} \frac{\tan^4 x}{\sin^2 x - x^2} = \lim_{x \rightarrow 0} \frac{x^4 + o(x^4)}{-\frac{x^4}{3} + o(x^4)} = -3$

$\sin^2 x \sim (x - \frac{x^3}{6})^2 + o(x^4) = x^2 - \frac{x^4}{3} + o(x^4)$

$\tan^4 x \sim [x + \frac{x^3}{3} + o(x^3)]^4 \sim x^4 + o(x^4)$

$\lim_{x \rightarrow 0} \frac{x^4}{(\sin x - x)(\sin x + x)} = \lim_{x \rightarrow 0} \frac{x^4}{-\frac{x^3}{6} \cdot 2x} = -3$

(2022-17) 当 $x \rightarrow 0^+$ 时, $\cos x - e^{-\frac{x}{2}} \sim ax^b$

$$\begin{aligned} \cos x - e^{-\frac{x}{2}} &\sim -\frac{1}{12}x^4 \\ \cos x &\sim 1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^4) \\ e^{-\frac{x}{2}} &\sim 1 - \frac{x}{2} + \frac{x^2}{8} + o(x^2) \\ e^x &\sim 1 + x + \frac{x^2}{2} + o(x^2) \end{aligned}$$

$\Rightarrow a = -\frac{1}{12} \quad b = 4$

→ 归结原则

8 归结原则

$$\lim_{x \rightarrow x_0} f(x) = A \iff \begin{array}{l} \text{任何} \\ |x_n| \rightarrow x_0 \end{array} \begin{array}{l} \lim_{n \rightarrow \infty} f(x_n) = A \\ \lim_{n \rightarrow \infty} y_n = A \end{array} \quad \begin{array}{l} \text{函数极限} \\ \text{数列极限} \end{array}$$

① $f(x)$ 与 x_n, y_n $\lim_{n \rightarrow \infty} f(x_n) \neq \lim_{n \rightarrow \infty} f(y_n)$ 右 → 左 取2个不同数列来否定原极限存在

eg $f(x) = x \sin \frac{1}{x}$ 取 $x_n = \frac{1}{n\pi}$ $\lim_{n \rightarrow \infty} n\pi \sin n\pi = 0$
 求 $\lim_{x \rightarrow 0} x \sin \frac{1}{x}$ 取 $x_n = \frac{1}{2m\pi + \frac{\pi}{2}}$ $\lim_{m \rightarrow \infty} (2m\pi + \frac{\pi}{2}) \sin(2m\pi + \frac{\pi}{2}) = \infty$ $\left\{ \begin{array}{l} \lim_{x \rightarrow 0} f(x) \text{ 不存在} \end{array} \right.$

② $\lim_{x \rightarrow x_0} f(x) = A \Rightarrow \lim_{n \rightarrow \infty} f(x_n) = A$ 左 → 右

常考: $\lim_{x \rightarrow 0} f(x) = A \rightarrow \lim_{n \rightarrow \infty} f(\frac{1}{n}) = A$

例 3.15

$$\lim_{n \rightarrow \infty} (n \tan \frac{1}{n})^{n^2} = \lim_{x \rightarrow \infty} \left(\frac{\tan x}{x} \right)^{\frac{1}{x^2}} \stackrel{1^\infty}{=} \lim_{x \rightarrow \infty} e^{\frac{1}{x^2}(\frac{\tan x}{x} - 1)} = e^{\frac{1}{3}}$$

→ 无穷小比较

9 无穷小比较

① 0 是特殊的无穷小 (最小的无穷小)

② $\lim_{x \rightarrow 0} \frac{\alpha(x)}{\beta(x)} = \begin{cases} 0 & \alpha \text{ 是比 } \beta \text{ 高阶的无穷小} \\ \infty & \alpha \text{ 是比 } \beta \text{ 低阶的无穷小} \\ C (C \neq 0) & \text{同阶无穷小} \\ 1 & \text{等价无穷小 } \alpha(x) \sim \beta(x) \end{cases}$

$\lim_{x \rightarrow 0} \left(\frac{\alpha(x)}{\beta(x)} \right)^k = 1$ α 是 β 的 k 阶无穷小

③ 并不是任意两个无穷小都可以比较

$x \rightarrow 0$ 时 $x \sin \frac{1}{x} = 0$. $\lim_{x \rightarrow 0} \frac{x \sin \frac{1}{x}}{x^2} = \lim_{x \rightarrow 0} \frac{1}{x} \sin \frac{1}{x}$ 不存在

(④ 无穷小运算规则 与 运算)

无穷小 $\times$ 有界 = 无穷小

$$⑤ \quad a^x - 1 \sim x \ln a \quad (1+x)^a - 1 \sim ax$$

函数的连续与间断

= 函数的连续与间断

1. 连续的定义: $\lim_{x \rightarrow x_0} f(x) = f(x_0)$ ★

2. 间断 $\lim_{x \rightarrow x_0^-} f(x) = \lim_{x \rightarrow x_0^+} f(x) = f(x_0)$

破坏②满足①: 可去间断点、
破坏①: 跳跃间断点、
 $\lim_{x \rightarrow x_0} f(x) = \infty$: 无穷间断点、
eg $\lim_{x \rightarrow 0} \sin \frac{1}{x}$: 振荡间断点、
...

例1.3.1

已知 $\lim_{x \rightarrow 1} f(x)$ 存在且 $f(x) = x^2 + x - 2 \lim_{x \rightarrow 1} f(x)$ 则 $\lim_{x \rightarrow 1} f(x) = ?$

$$\lim_{x \rightarrow 1} f(x) = A \quad \lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} (x^2 + x - 2A) = A$$

$$\Rightarrow \lim_{x \rightarrow 1} x^2 + x = 2A = 2 \Rightarrow \lim_{x \rightarrow 1} f(x) = A = \frac{2}{3}$$

$\Rightarrow \lim_{x \rightarrow 1} f(x) = A, f'(x_0) = B, \int_a^b f(x) dx = C, f'_x(x_0, y_0) = D, \int_0^b f(x, y) dy = F$ 是常数

例1.3.2

当 $X \rightarrow 1$ 时, 函数 $\frac{x-1}{x-1} e^{\frac{1}{x-1}}$ 的极限

$$\lim_{x \rightarrow 1^+} \frac{x-1}{x-1} e^{\frac{1}{x-1}} = \lim_{t \rightarrow 0^+} (t+2) e^{\frac{1}{t}} = \infty$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{t \rightarrow 0^-} (t+2) e^{\frac{1}{t}} = 0 \quad \text{不存在且不} \infty$$

$$\begin{array}{l} X \rightarrow \infty \\ \left\{ \begin{array}{l} X \rightarrow +\infty \\ X \rightarrow -\infty \end{array} \right. \end{array} \quad \begin{array}{l} X \rightarrow x_0 \\ \left\{ \begin{array}{l} X \rightarrow x_0^+ \\ X \rightarrow x_0^- \end{array} \right. \end{array}$$

$$1) \lim_{x \rightarrow \infty} e^x = +\infty \rightarrow +\infty \quad -\infty \rightarrow 0 \quad 3) \lim_{x \rightarrow \frac{\pi}{2}} \arctan x: +\infty \rightarrow \frac{\pi}{2} \quad -\infty \rightarrow -\frac{\pi}{2}$$

$$2) \lim_{x \rightarrow 0} \frac{\sin x}{|x|} = 0^+ \rightarrow 1 \quad 0^- \rightarrow -1 \quad 4) \lim_{x \rightarrow 0} [x]: 0^+ \rightarrow 0 \quad 0^- \rightarrow -1$$

$\Rightarrow$ 命 $\lim_{x \rightarrow 0} [k \cdot \arctan x + \frac{e^x - \pi}{e^x + 1}]$ 存在

求 k 并计算此极限

$$\begin{aligned} \lim_{x \rightarrow 0^+} k \arctan x &= \frac{\pi}{2} \cdot k & \lim_{x \rightarrow 0^+} f(x) &= \frac{\pi}{2} k \\ \lim_{x \rightarrow 0^+} \frac{e^x - \pi}{(e^x)^2 + 1} &= 0 & & \\ \lim_{x \rightarrow 0^-} k \arctan x &= -\frac{\pi}{2} \cdot k & \lim_{x \rightarrow 0^-} f(x) &= -\frac{\pi}{2} k - \pi \\ \lim_{x \rightarrow 0^-} \frac{e^x - \pi}{(e^x)^2 + 1} &= -\pi & & \end{aligned}$$

$$\Rightarrow \frac{\pi}{2} k = -\frac{\pi}{2} k - \pi \Rightarrow k = -1 \quad \lim_{x \rightarrow 0} f(x) = -\frac{\pi}{2}$$

例 1.3.3 局部有界性

在下列区间中, $f(x) = \frac{x \sin(x-3)}{(x-1)(x-3)^2}$ 有界的是 ()

A $(-2, 1)$ B $(-1, 0)$ C $(1, 2)$ D $(2, 3)$

无穷间断点: $x=1$ $x=3 \rightarrow$ B.

$f(x)$ 在 $[a, b]$ 上连续 $\Rightarrow f(x)$ 在 $[a, b]$ 有界

$f(x)$ 在 (a, b) 上连续: $f(x)$ 在 (a, b) 有界

$\lim_{x \rightarrow a^+} f(x)$ 存在 $(a, a+\delta)$ 有界

$\lim_{x \rightarrow b^-} f(x)$ 存在 $(b-\delta, b)$ 有界

例 1.3.4

$$\lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{x - e^{\frac{1}{2x}} + 1} = \lim_{t \rightarrow 0^+} \frac{t}{t^2 - e^{\frac{1}{2t}} + 1} = \lim_{t \rightarrow 0^+} \frac{t}{-2t - t^2} = -\frac{1}{2}$$

$$e^{\frac{1}{2t}} \sim 1 + \frac{1}{2t} + \frac{1}{8t^2} + o(\frac{1}{t^2})$$

七种未定式 (I) $\frac{0}{0}, \frac{\infty}{\infty}, \infty \cdot 0$

(II) $0 - \infty$

(III) $\infty^0, 0^0, 1^\infty$

$$\ln(1+x) \sim x - \frac{x^2}{2} + \frac{x^3}{3}$$

例 1.3.5

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+\tan x} - \sqrt{1+\sin x}}{x \ln(1+x) - x^2} = \lim_{x \rightarrow 0} \frac{(1+\tan x) - (1+\sin x)}{[x \ln(1+x) - x^2](\sqrt{1+\tan x} + \sqrt{1+\sin x})} = \lim_{x \rightarrow 0} \frac{\frac{1}{2}x^3}{\underbrace{[x \ln(1+x) - x^2]}_{-\frac{x^2}{2} + o(x^2)} \underbrace{(\sqrt{1+\tan x} + \sqrt{1+\sin x})}_2} = -\frac{1}{2}$$

例 1.3.6

$$(1+x)^a \sim 1 + ax + \frac{a(a-1)}{2}x^2 + o(x^2)$$

$$1) \lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2+x-1} + x+1}{\sqrt{x^2 + \sin x}} = \lim_{x \rightarrow -\infty} \frac{\sqrt{4+\frac{1}{x}-\frac{1}{x^2}} - 1 - \frac{1}{x}}{\sqrt{1+\frac{\sin x}{x^2}}} = \lim_{x \rightarrow -\infty} \frac{\sqrt{4+x-x^2} - 1 - x}{\sqrt{1+x^2 \sin \frac{1}{x}}} = \lim_{x \rightarrow -\infty} \frac{(1+\frac{x-x^2}{4}+o(x)) - 1 - x}{1+\frac{1}{2} \sin \frac{1}{x} + o(x^2)} = 1$$

$$2) \lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2+x-1} + x+1}{\sqrt{x^2 + \sin x}} = \lim_{t \rightarrow +\infty} \frac{\sqrt{4t^2-t-1} - t+1}{\sqrt{t^2 - \sin t}} = \lim_{t \rightarrow +\infty} \frac{2t^2 + t - 2}{\sqrt{t^2 - \sin t} (\sqrt{4t^2-t-1} + |t^2-2t+1|)}$$

$$= \lim_{t \rightarrow +\infty} \frac{2 + \frac{1}{t} - \frac{2}{t^2}}{\sqrt{1 - \frac{\sin t}{t^2}} (\sqrt{4 - \frac{1}{t} - \frac{1}{t^2}} + \sqrt{1 - \frac{2}{t} + \frac{1}{t^2}})} = 1$$

例 1.3.7

$$\lim_{x \rightarrow -\infty} x(\sqrt{x^2+100} + x) = \lim_{t \rightarrow +\infty} -t(\sqrt{t^2+100} - t) = \lim_{t \rightarrow +\infty} \frac{-t(t^2+100-t^2)}{\sqrt{t^2+100} + t} = -50$$

例 1.3.8

$$\lim_{x \rightarrow 1^-} \ln x \ln(1-x) \stackrel{x=1-t}{=} \lim_{t \rightarrow 0^+} \ln(t+1) \ln(-t) \stackrel{u=-t}{=} \lim_{u \rightarrow 0^+} \ln(1-u) \ln u = \lim_{u \rightarrow 0^+} -u \ln u = 0$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + o(x^3)$$

例 1.3.9

$$D' \text{ 算 } \lim_{x \rightarrow 0} \frac{\arcsin x - \arctan x}{\sin x - \tan x} = -1$$

例 1.3.10

求 $I = \lim_{x \rightarrow 0} x \left[\frac{10}{x} \right]$, 其中 $[\cdot]$ 为取整符号

用 " $x-1 < [x] \leq x$ "

$$\frac{10}{x} - 1 < \left[\frac{10}{x} \right] \leq \frac{10}{x}$$

$$\Rightarrow x > 0 \text{ 时 } 10 - x < x \left[\frac{10}{x} \right] \leq 10$$

$$x < 0 \text{ 时 } 10 - x > x \left[\frac{10}{x} \right] \geq 10$$

$$\rightarrow 10 \leftarrow$$

$$\ln(1+x) \sim x - \frac{x^2}{2} + \frac{x^3}{3} + o(x^3)$$

例 1.3.11

$$\lim_{x \rightarrow 0} \left[\frac{1}{\ln(1+x)} - \frac{1}{x} \right] = \lim_{x \rightarrow 0} \frac{x - \ln(1+x)}{x \ln(1+x)} = \lim_{x \rightarrow 0} \frac{\frac{x^2}{2}}{x^2} = \frac{1}{2}$$

$x = \frac{1}{t}$

有分母则通分, 没有分母(创造分母)再通分.

例 1.3.12

$$\lim_{x \rightarrow +\infty} [x^2(e^{\frac{1}{x}} - 1) - x] = \lim_{x \rightarrow +\infty} \frac{x(e^{\frac{1}{x}} - 1) - 1}{1/x} = \lim_{t \rightarrow 0^+} \frac{e^t - 1 - t}{t^2} = \frac{1}{2}$$

$e^t \sim 1 + t + \frac{t^2}{2}$

★ 例 1.3.13

$$\lim_{x \rightarrow +\infty} (x + \sqrt{1+x^2})^{\frac{1}{x}} = \lim_{x \rightarrow +\infty} e^{\frac{1}{x} \ln(x + \sqrt{1+x^2})} = \lim_{x \rightarrow +\infty} e^{\frac{\frac{1}{1+x^2}}{\frac{1}{x}}} = 1$$

$$(\ln(x + \sqrt{1+x^2}))' = \frac{1}{1+x^2}$$

例 1.3.14

$$\lim_{x \rightarrow 0} \left(\frac{e^x + e^{2x} + e^{3x}}{3} \right)^{\frac{1}{x}} \stackrel{1}{=} \lim_{x \rightarrow 0} e^{\frac{e^x + e^{2x} + e^{3x}}{3} \cdot \frac{1}{x}} = \lim_{x \rightarrow 0} e^{\frac{e^x - 1}{x} + \frac{e^{2x} - 1}{2x} + \frac{e^{3x} - 1}{3x}} = \lim_{x \rightarrow 0} e^{22} = e^{22}$$

例 1.3.15 归结原则

$$\lim_{n \rightarrow \infty} (n \tan \frac{1}{n})^{n^2} \quad (n \text{ 为正整数}) \stackrel{x=1/n}{=} \lim_{x \rightarrow 0^+} \left(\frac{\tan x}{x} \right)^{\frac{1}{x^2}} = \lim_{x \rightarrow 0^+} e^{\frac{1}{x^2} \left(\frac{\tan x - x}{x} \right)} = e^{\frac{1}{3}}$$

例 1.3.17 已知极限反求参数

$$\text{设 } \lim_{x \rightarrow 0} \frac{\ln(1+x) - (ax+bx^2)}{x^2} = 2 \quad \text{求 } a, b$$

$$\ln(1+x) = x - \frac{x^2}{2} + o(x^2), \quad \lim_{x \rightarrow 0} \frac{\ln(1+x) - (ax+bx^2)}{x^2} = \lim_{x \rightarrow 0} \frac{(1-a)x - (\frac{1}{2}+b)x^2}{x^2} = 2$$

$$\Rightarrow a=1 \quad b=-\frac{5}{2}$$

例 1.3.18 无穷小比较

$x \rightarrow 0$ 时 $e^x - (ax^2 + bx + 1)$ 是比 x^2 高阶的无穷小, 则

$$e^x = 1 + x + \frac{x^2}{2} + o(x^2) \Rightarrow a=1 \quad b=\frac{1}{2}$$

例 1.3.19 无穷小比较

$x \rightarrow 0$ 时 $e^{\tan x} - e^x$ 与 x^n 是同阶无穷小, $n=?$

$$\begin{aligned} e^x &\sim 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3) \\ e^{\tan x} &\sim 1 + \tan x + \frac{(\tan x)^2}{2} + \frac{(\tan x)^3}{6} + o(x^3) \sim 1 + x + \frac{x^3}{3} + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3) \\ \tan x &\sim x + \frac{x^3}{3} + o(x^3) \Rightarrow n=3 \end{aligned}$$

$$\lim_{x \rightarrow 0} e^{\tan x} - e^x = \lim_{x \rightarrow 0} e^x (e^{\tan x - x} - 1) = \lim_{x \rightarrow 0} e^x (\tan x - x)^{\frac{1}{3}} = \lim_{x \rightarrow 0} \frac{x^3}{3} \Rightarrow n=3$$

$$e^{f(x)} - e^{g(x)} = e^{g(x)} [e^{f(x)-g(x)} - 1] \sim f(x) - g(x) \quad (f(x) \rightarrow 0, g(x) \rightarrow 0)$$

例 1.3.20 无穷小比较

设 $a_1 = x(\cos x - 1)$ $a_2 = \sqrt{x} \ln(1 + \sqrt{x})$ $a_3 = \sqrt[3]{x+1} - 1$ $x \rightarrow 0$ 时 a_1, a_2, a_3 的高阶无穷小

$$a_1 = x(\cos x - 1) \sim x \cdot (-\frac{x^2}{2}) \sim o(x^2)$$

$$a_2 = \sqrt{x} \ln(1 + \sqrt{x}) \sim \sqrt{x} \cdot \sqrt{x} \sim o(x^{\frac{1}{2}})$$

$$a_3 = \sqrt[3]{x+1} - 1 \sim \frac{1}{3}x \sim o(x)$$

a_1, a_2, a_3

例 1.3.21 函数的连续与间断

$$f(x) = \begin{cases} x+a, & x \leq 0 \\ e^x(\sin x + \cos x), & x > 0 \end{cases} \quad \text{在 } (-\infty, +\infty) \text{ 连续 } a=?$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} e^x(\sin x + \cos x) = 1 = f(0) = a \Rightarrow a=1$$

讨论间断点，只看无定义点，与分段函数分段点。

例 1.3.22

$$f(x) = \frac{\ln|x|}{|x-1|} \sin x$$

无定式: $x=1$ $\lim_{x \rightarrow 1} \frac{\ln|x|}{|x-1|} \sin x = \lim_{t \rightarrow 0} \frac{\ln(1+t)}{|1+t|} \sin 1$

$$x=0 \quad \lim_{x \rightarrow 0} \sin x \ln|x| = 0 \quad \text{可去}$$

$$\lim_{t \rightarrow 0^+} \frac{\ln(1+t)}{1+t} = 1$$

$$\lim_{t \rightarrow 0^-} \frac{\ln(1+t)}{-t} = -1$$

跳跃间断点

例 1.3.23

$$f(x) = \frac{x^2 - x}{x^2 - 1} \sqrt{1 + \frac{1}{x^2}}$$

$$= \frac{x(x-1)}{(x-1)(x+1)} \sqrt{\frac{x^2+1}{x^2}} \quad x \neq 0$$

$x=1$: 可去, $x=-1$: 无穷, $x=0$: 可去

例 1.3.24

$$\lim_{t \rightarrow x} \left(\frac{\sin t}{\sin x} \right)^{\frac{x}{\sin t - \sin x}} = f(x)$$

$$t=x+u \quad \lim_{\substack{t \rightarrow x \\ u \rightarrow 0}} \left(\frac{\sin(x+u)}{\sin x} \right)^{\frac{x}{\sin(x+u) - \sin x}} \stackrel{1^\infty}{=} \lim_{u \rightarrow 0} e^{\frac{x}{\sin(x+u) - \sin x} \cdot \frac{\sin(x+u) - \sin x}{\sin x}} = e^{\frac{x}{\sin x}}$$

间断点, $\sin x = 0 \Rightarrow x = k\pi$ $x=0$ 是可去间断点, 其他是无穷间断点

习 1.3.1

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + o(x^3) \quad e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + o(x^6)$$

$$\lim_{x \rightarrow 0} \frac{e^{-\frac{1}{x^2}}}{x^{100}} = \lim_{x \rightarrow 0} \frac{e^{-\frac{1}{x^2}} \cdot 2x^{-3}}{2 \cdot x^{97}} = \lim_{x \rightarrow 0} \frac{e^{-\frac{1}{x^2}} \cdot x^{-3}}{\frac{2x^2}{97} \cdot x^{95}} = \lim_{x \rightarrow 0} \frac{e^{-\frac{1}{x^2}}}{\frac{2x^2}{97} \cdot x^{96}}$$

$$= \lim_{x \rightarrow 0} \frac{2 \cdot 2 \cdots 2 \cdot 2 \cdot e^{-\frac{1}{x^2}}}{2 \cdot 4 \cdot 6 \cdots 96 \cdot 98 \cdot x^2} = \frac{1}{44!} \lim_{x \rightarrow 0} \frac{1}{x^2} e^{-\frac{1}{x^2}} = \frac{1}{44!} \lim_{t \rightarrow +\infty} t \cdot e^{-t} = 0$$

$$\frac{1}{x^2} = t \quad \lim_{t \rightarrow +\infty} t^{100} e^{-t^2} = \lim_{u \rightarrow +\infty} \frac{u^{50}}{e^u} = 0$$

习 1.3.2

$$I = \lim_{x \rightarrow 0} \left(\frac{\ln(1+e^{\frac{2}{x}})}{\ln(1+e^{\frac{1}{x}})} + a[x] \right) \quad I = ? \quad a = ?$$

$$\lim_{t \rightarrow +\infty} \frac{\ln(1+e^{2t})}{\ln(1+e^t)} + 0 = \lim_{t \rightarrow +\infty} \frac{(1+e^{2t})e^{2t} \cdot 2}{(1+e^t)e^t} = 2$$

$$\lim_{t \rightarrow +\infty} \frac{\ln(1+e^{2t})}{\ln(1+e^t)} - a = \lim_{t \rightarrow +\infty} \frac{(1+e^{2t})e^{2t} \cdot 2}{(1+e^t)e^t} - a = -a$$

$$\left. \begin{array}{l} \lim_{t \rightarrow +\infty} \frac{\ln(1+e^{2t})}{\ln(1+e^t)} + 0 = 2 \\ \lim_{t \rightarrow +\infty} \frac{\ln(1+e^{2t})}{\ln(1+e^t)} - a = -a \end{array} \right\} \rightarrow a = -2 \quad I = 2$$

习 1.3.3

已知 $a > 0, b > 0$ 求 $\lim_{x \rightarrow +\infty} x(a^{\frac{1}{x}} - b^{\frac{1}{x}}) = ?$

$$\lim_{x \rightarrow +\infty} x(a^{\frac{1}{x}} - b^{\frac{1}{x}}) \stackrel{0}{=} \lim_{t \rightarrow 0^+} \frac{a^t - b^t}{t} = \lim_{t \rightarrow 0^+} \frac{e^{t \ln a} - e^{t \ln b}}{t}$$

$$= \lim_{t \rightarrow 0^+} \frac{e^{t \ln b} (e^{t \ln \frac{a}{b}} - 1)}{t} = \frac{e^{t \ln b} \cdot t \cdot \ln \frac{a}{b}}{t} = \ln \frac{a}{b}$$

$$e^x(x^2+x-1) - 1 \quad e^x \sim 1+x+\frac{x^2}{2}$$

习 1.3.4

$$\lim_{x \rightarrow 0} \left(\frac{e^x + x e^x}{e^x - 1} - \frac{1}{x} \right) = \lim_{x \rightarrow 0} \frac{x e^x + x^2 e^x - e^x + 1}{x(e^x - 1)} = \lim_{x \rightarrow 0} \frac{(1+x+\frac{x^2}{2})(x^2+x-1)+1}{x^2} = 1.5$$

习1.3.5

设 $a \neq \pm 1$, 计算 $\lim_{n \rightarrow \infty} \ln \left[\frac{n-2na+1}{n(1-2a)} \right]^n = \lim_{k \rightarrow 0} \frac{\ln \left[\frac{1-2a+k}{1-2a} \right]}{\frac{k}{n}} = \lim_{k \rightarrow 0} \frac{\ln \left[1 + \frac{k}{1-2a} \right]}{\frac{k}{n}} = \frac{1}{1-2a}$

习1.3.6

$$\begin{aligned} ① \lim_{x \rightarrow 0} \frac{x^x - (\sin x)^x}{x^2 \ln(1+x)} &= \lim_{x \rightarrow 0} \frac{e^{x \ln x} - e^{x \ln \sin x}}{x^2} = \lim_{x \rightarrow 0} \frac{x(\ln x - \ln \sin x)}{x^3} = \lim_{x \rightarrow 0} \frac{\ln \frac{x}{\sin x}}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{\sin x \cdot (\sin x - x \cos x)}{2x^2 \sin^2 x} = \lim_{x \rightarrow 0} \frac{\cos x (\tan x - x)}{2x^3} = \frac{1}{6} \\ ② \lim_{x \rightarrow 0^+} \frac{x^x - (\sin x)^x}{x^2 \ln(1+x)} &= - \lim_{x \rightarrow 0^+} \frac{x^x \left[\left(\frac{\sin x}{x} \right)^x - 1 \right]}{x^2} \stackrel{\lim_{x \rightarrow 0} x^x = 1}{=} - \lim_{x \rightarrow 0^+} \frac{e^{x \ln \frac{\sin x}{x}} - 1}{x^2} \end{aligned}$$

习1.3.7

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{(a_1^x + a_2^x + \dots + a_n^x)}{n} \right)^{\frac{1}{x}} &= \lim_{x \rightarrow 0} \left(1 + \frac{(a_1^x - 1) + (a_2^x - 1) + \dots + (a_n^x - 1)}{n} \right)^{\frac{1}{x}} \\ &= \lim_{x \rightarrow 0} \left[\left(1 + \frac{(a_1^x - 1) + (a_2^x - 1) + \dots + (a_n^x - 1)}{n} \right)^{\frac{n}{(a_1^x - 1) + (a_2^x - 1) + \dots + (a_n^x - 1)}} \right]^{\frac{(a_1^x - 1) + (a_2^x - 1) + \dots + (a_n^x - 1)}{x}} \\ &= \lim_{x \rightarrow 0} e^{\frac{(a_1^x - 1) + (a_2^x - 1) + \dots + (a_n^x - 1)}{x}} = \lim_{x \rightarrow 0} e^{\frac{x(\ln a_1 + \ln a_2 + \dots + \ln a_n)}{x}} = e^{a_1 a_2 \dots a_n} \end{aligned}$$

习1.3.8

证 $\lim_{x \rightarrow 0} \frac{\ln \left[1 + \frac{f(x)}{\sin x} \right]}{2^x - 1} = 2$ 求 $\lim_{x \rightarrow 0} \frac{f(x)}{x^2}$

$$\lim_{x \rightarrow 0} \frac{\ln \left[1 + \frac{f(x)}{\sin x} \right]}{x} = 2 \ln 2, \quad \lim_{x \rightarrow 0} \frac{\frac{f(x)}{\sin x} \cdot x}{x^2} = 2 \ln 2, \quad \lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 2 \ln 2$$

习1.3.9

$f(x)$ 在 $(-\infty, +\infty)$ 内有定义, 且 $\lim_{x \rightarrow \infty} f(x) = a$, $g_{\infty} = \begin{cases} f(x), & x \neq 0 \\ 0, & x = 0 \end{cases}$

$$x=0 \quad g(0)=0 \quad \lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow \infty} f(x) = a$$

$a=0$ 时, 连续 $a \neq 0$ 时 跳跃间断点 (1)

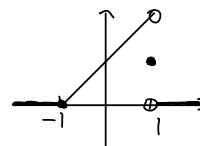
例 3.10

$$f(x) = \lim_{n \rightarrow \infty} \frac{1+x}{1+x^{2n}} \quad \text{间断点}$$

$$x \in (-\infty, -1) \cup (1, +\infty) \quad \text{时} \quad f(x) = 0$$

$$x \in (-1, 1) \quad \text{时} \quad f(x) = 1+x$$

$$x = \pm 1 \quad \text{时} \quad f(x) = \frac{1 \pm 1}{2}$$



(B)

300.3.1 (B)

$$A. \lim_{x \rightarrow \pi} \frac{\sin x}{x} \stackrel{t=x-\pi}{=} \lim_{x \rightarrow 0} \frac{-\sin x}{x+\pi} = 0$$

$$B. \lim_{x \rightarrow \infty} x \sin \frac{1}{x} \stackrel{t=\frac{1}{x}}{=} \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$C. \lim_{x \rightarrow \infty} \frac{1}{x} \sin \frac{1}{x} = \lim_{x \rightarrow 0} x \sin x = 0$$

$$D. \lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$$

300.3.2 (D)

$$\lim_{x \rightarrow 1} 2e^{\frac{1}{x-1}} \stackrel{t=x-1}{=} \lim_{x \rightarrow 0} 2e^{\frac{1}{x}} \quad \text{不存在, 但不为 } \infty$$

300.3.3

$$e^{-\sqrt{x}} - 1 \sim -\sqrt{x}$$

$$\frac{\sin x}{\sqrt{x}} \sim \frac{x}{\sqrt{x}} = \sqrt{x} \quad (B)$$

$$\ln(1+x) = \frac{1}{2} \ln(1+x) \sim \frac{x}{2}$$

$$\ln(1-\sqrt{x}) \sim -\sqrt{x}$$

$$x \frac{e^{\frac{1}{x-1}} + 1}{x-1}$$

300.3.4

$$\lim_{x \rightarrow \infty} \left(\frac{x+2}{x-1} \right)^x = \lim_{x \rightarrow \infty} e^{x \frac{(x+2)-(x-1)}{x-1}} = \lim_{x \rightarrow \infty} e^{\frac{3x}{x-1}} = e^3$$

300.3.5

$$\lim_{x \rightarrow \infty} \frac{2e^{\frac{1}{x}} + e^{-\frac{1}{x}}}{e^{\frac{1}{x}} - e^{-\frac{1}{x}}} \stackrel{x=t}{=} \lim_{t \rightarrow \infty} \frac{2e^t + e^{-t}}{e^t - e^{-t}} = 0$$

300.3.6

$$\lim_{x \rightarrow 0} \frac{\sin x + x^2 \sin \frac{1}{x}}{(2+x^2) \ln(1+x)} = \lim_{x \rightarrow 0} \frac{x - \frac{x^3}{6} + o(x^2)}{2x} = \frac{1}{2}$$

300.3.7

$x \rightarrow 0$ $\sqrt{1+ax^2} - 1$ 与 $\sin^2 x$ 等价无穷小, 求 a .

$$\lim_{x \rightarrow 0} \frac{(1+ax^2)^{\frac{1}{2}} - 1}{\sin^2 x} = \lim_{x \rightarrow 0} \frac{\frac{a}{2}x^2}{x^2} = 1 \Rightarrow a=2$$

300.3.8

$x \rightarrow 0^+$ 时, 若 $\sin x^a$, $(1-\cos x)^{\frac{1}{a}}$ 均是比 x 高阶的无穷小量, 求 a 范围

$$\sin x^a \sim x^a \quad a > 1$$

$$(1-\cos x)^{\frac{1}{a}} \sim \left(\frac{x^2}{2}\right)^{\frac{1}{a}} = \frac{1}{2^{\frac{1}{a}}} x^{\frac{2}{a}} \quad \frac{2}{a} > 1 \quad a < 2$$

$$a \in (1, 2)$$

300.3.9

$$f(x) = \frac{\sin(x-1)}{x^2-1} = \frac{\sin(x-1)}{(x-1)(x+1)}$$

$x=1$ 可去 $x=-1$ 无穷

300.3.10

$$f(x) = \begin{cases} \frac{1-e^{\sin x}}{\arcsin x}, & x > 0 \\ a, & x \leq 0 \end{cases} \quad \text{在 } x=0 \text{ 处连续. } a=?$$

$$\lim_{x \rightarrow 0^+} \frac{1-e^{\sin x}}{\arcsin x} = \lim_{x \rightarrow 0^+} \frac{-x}{x} = -1 \quad a = -1$$

300.3.11

$$f(x) = \begin{cases} \frac{1-e^{\tan x}}{\arcsin \frac{x}{2}}, & x > 0 \\ ae^{2x}, & x \leq 0 \end{cases} \quad x=0 \text{ 连续}$$

$$\lim_{x \rightarrow 0^+} \frac{-x}{\frac{x}{2}} = -2 = f(0) = a \Rightarrow a = -2$$

300.3.12

$$f(x) = \frac{x^2-x}{(x^2-1)|x|} = \frac{x(x-1)}{(x-1)(x+1)|x|} \quad \begin{matrix} x=0 & \text{可去} & \text{跳跃} \\ x=1 & \text{可去} \\ x=-1 & \text{无穷} \end{matrix}$$

300.3.13

$$f(x) = \begin{cases} \frac{\ln(\cos(x-1))}{1-\sin \frac{\pi}{2}x}, & x \neq 1 \\ 1, & x = 1 \end{cases} \quad \lim_{x \rightarrow 1^+} f(x) = \lim_{\substack{t \rightarrow 0^+ \\ x=t+1}} \frac{\ln(\cos t)}{1-\sin(\frac{\pi}{2}t+\frac{\pi}{2})} = \lim_{t \rightarrow 0^+} \frac{\ln(\cos t)}{1-\cos(\frac{\pi}{2}t)}$$

$$= \lim_{t \rightarrow 0^+} \frac{-\sin t \sim t}{\sin(\frac{\pi}{2}t) \cdot \frac{\pi}{2} \cdot \cos t} = -\frac{4}{\pi^2} \rightarrow f(1) = -\frac{4}{\pi^2}$$